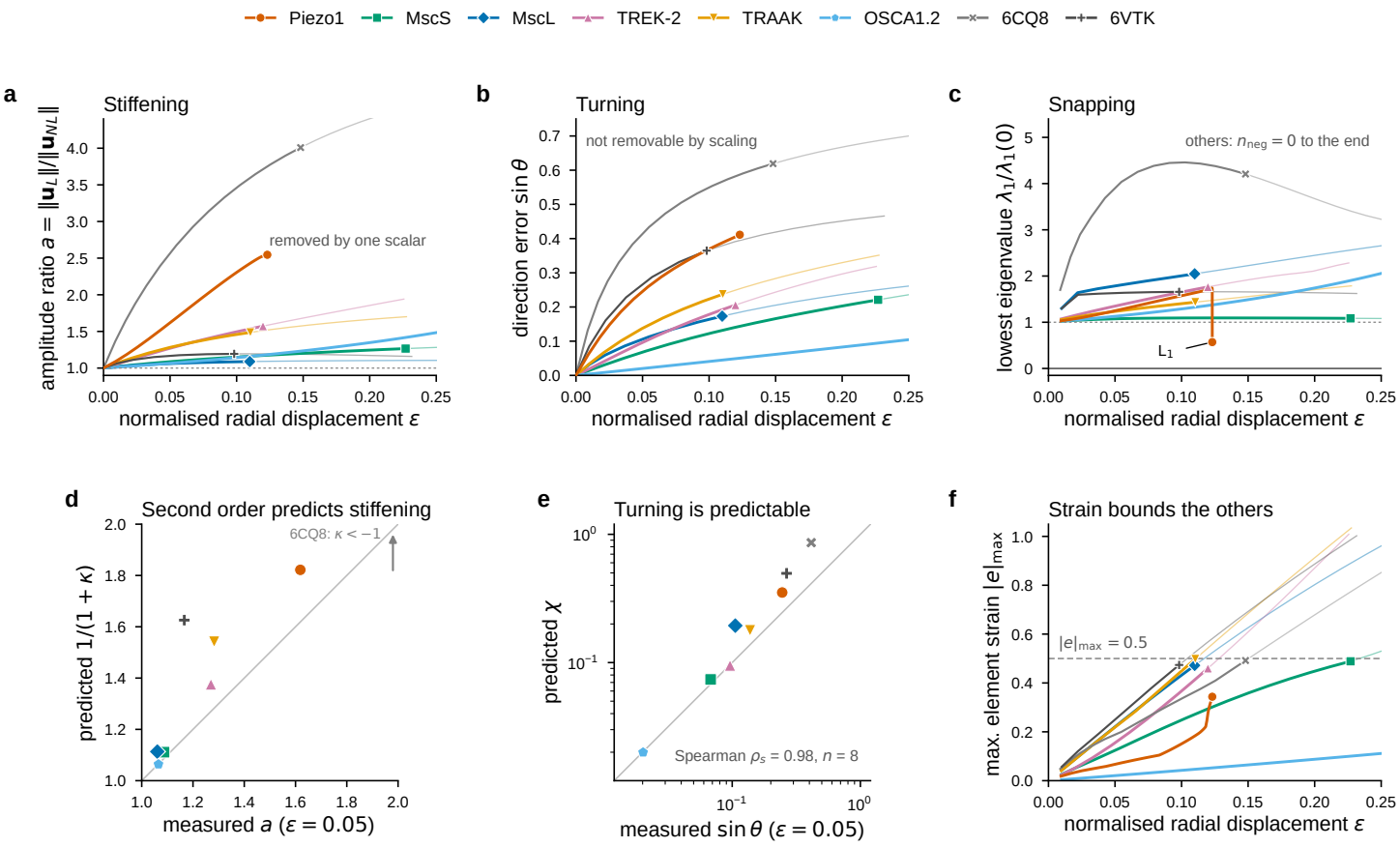
Piezo1
MscS
MscL
TREK-2
TRAAK
OSCA1.2
6CQ8
6VTK
a
Stiffening
amplitude ratio $a = \|\mathbf{u}_L\|/\|\mathbf{u}_{NL}\|$
removed by one scalar
normalised radial displacement ε
b
Turning
direction error $\sin\theta$
not removable by scaling
normalised radial displacement ε
c
Snapping
lowest eigenvalue $\lambda_1/\lambda_1(0)$
others: $n_{\text{neg}} = 0$ to the end
L_1
normalised radial displacement ε
d
Second order predicts stiffening
6CQ8: $\kappa < -1$
predicted $1/(1+\kappa)$
measured a ($\varepsilon = 0.05$)
e
Turning is predictable
predicted χ
Spearman $\rho_s = 0.98$, $n = 8$
measured $\sin\theta$ ($\varepsilon = 0.05$)
f
Strain bounds the others
max. element strain $|e|_{\max}$
$|e|_{\max} = 0.5$
normalised radial displacement ε